

1.2 River processes

Erosion

Erosion is the wearing away of the landscape. River erosion is a process that occurs as rivers flow over the land. There are four main types of river erosion:

- **abrasion**
- **hydraulic action**
- **attrition**
- **solution.**

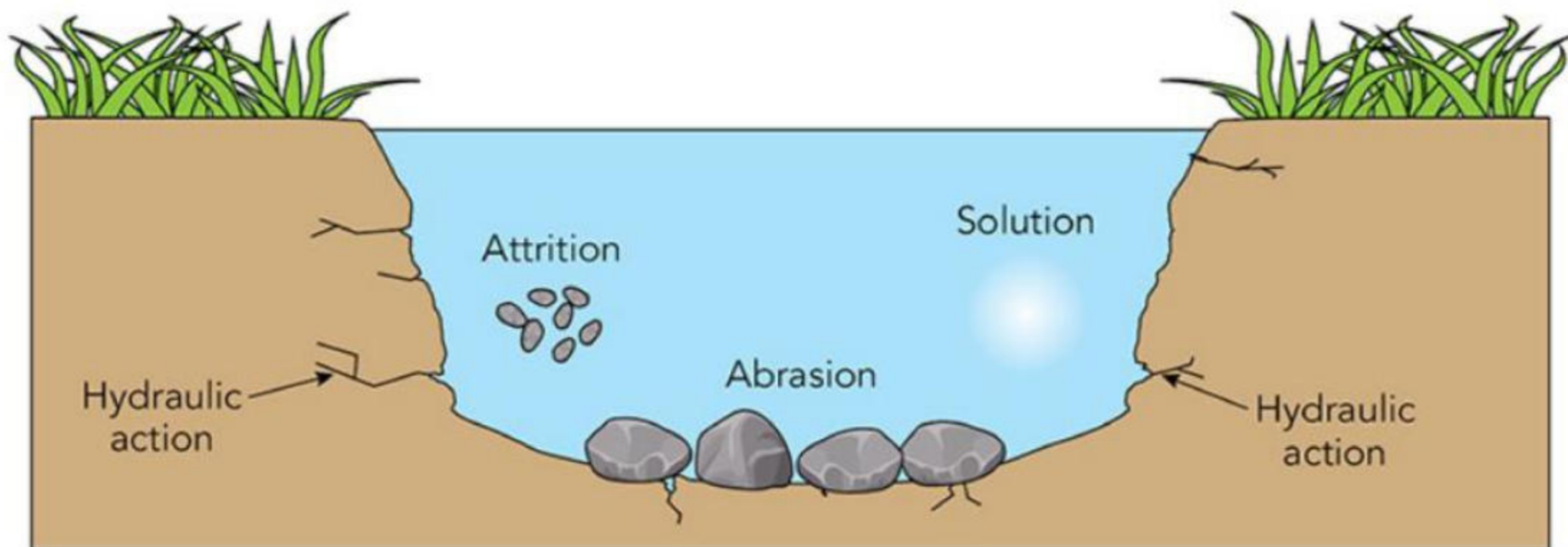


Figure 1.1: Types of river erosion

TASK 1

- Work in pairs to **create diagrams** to show how each type of erosion works. You might be creative about this, for example, the diagram for 'solution' could show sugar being dissolved in a cup of coffee.
- Share your diagrams with another pair in the class. Which pair created the best diagrams and why?



Source A: Rivers erode the landscape around them

The rate, or speed, of river erosion is influenced by several factors, including:

- the **gradient** of the river – steeper slopes lead to greater levels of energy, which generally leads to faster rates of erosion
- the size and type of sediments transported by the river – larger and more abrasive materials, such as sharp rocks, can increase rates of erosion
- the **geology**, or type of rock, that the river flows over – softer rocks, such as mudstone, erode more quickly than harder rocks, such as granite.

- a Suggest which types of river erosion could take place in this location.
- b Identify evidence within the image to show that the river is eroding the landscape.

Transportation

Transportation is the process by which rivers move eroded sediments and materials downstream. This happens in four main ways:

- Traction:** large boulders and rocks are pushed, dragged, or rolled along the riverbed by the force of flowing water. Traction usually takes place near the source of a river and it involves large rocks that are too heavy to be picked up. Moving bedload by traction requires a lot of energy.
- Saltation:** small pebbles and stones are lifted and carried by the flow of the river. They are too heavy to be suspended in the water for long periods of time, so they fall back down to the riverbed. This means they appear to bounce along. Saltation is more common closer to the source of a river, where bedload size is greater.
- Suspension:** fine sediment particles, such as silt and clay, are carried along by the river. These particles remain suspended (carried) in the water because they are small and light, which means they can easily be carried along. Suspension is more common closer to the mouth of a river, where bedload size is very small.
- Solution:** soluble minerals from rocks and sediments are dissolved within the water. These particles are so small that they require very little energy to transport and the river carries them over great distances. Solution typically takes place closer to the mouth of a river, where very fine material is found.

TASK 3

Create a copy of the table below. Add these words to the table in the correct places:

traction suspension very large small
saltation solution large fine

	Longer distances	Shorter distances
Type of transportation		
Size of bedload		

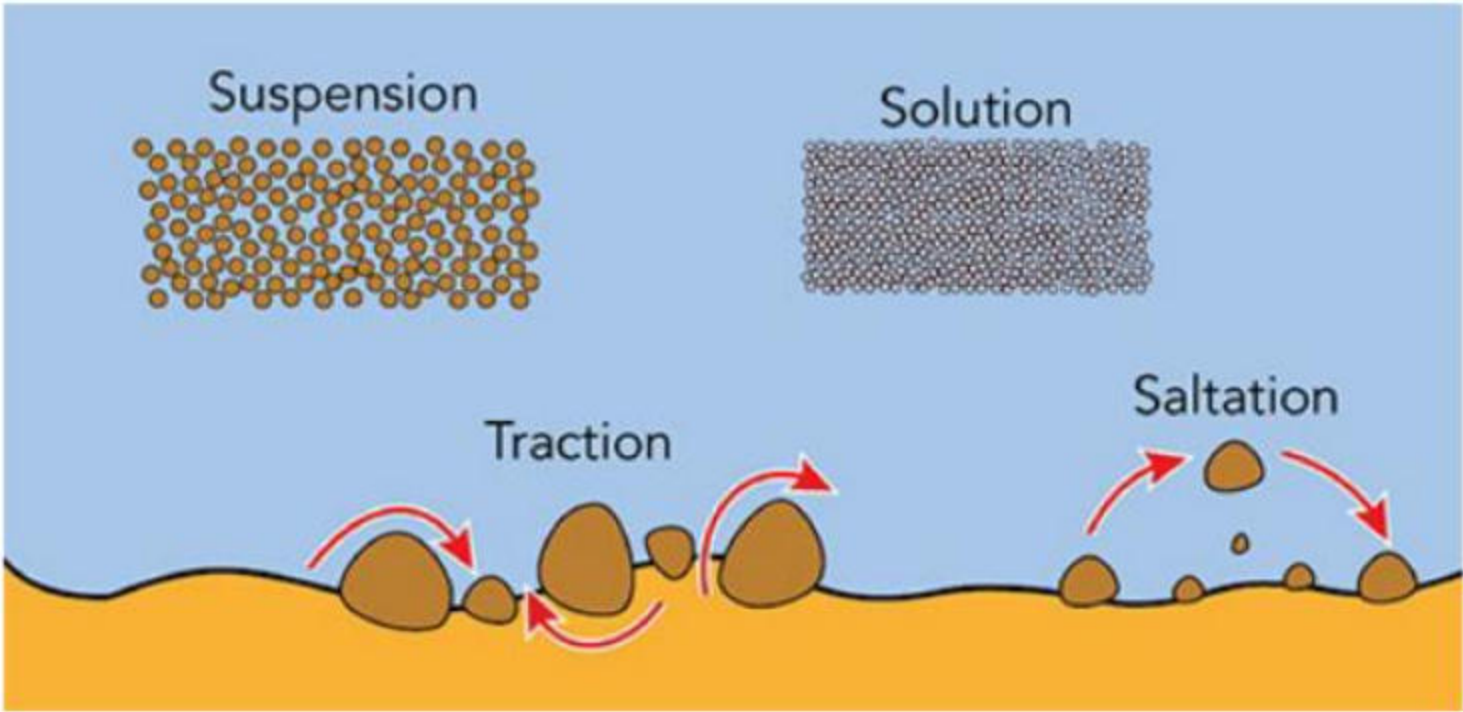


Figure 1.2: The processes of river transportation

TIP

Solution is a process of erosion and transportation. Within erosion, minerals and fine sediment are dissolved into a solution. Within transportation, minerals and fine sediment are moved in solution form.

REFLECTION

How did you decide where to place the words listed in Task 3? Did you check the text from the transportation section, or did you work out the answers by thinking about how transportation works? Understanding how processes work will help you to remember key facts.

Deposition

Deposition occurs when a river drops or deposits sediment it has been carrying. This takes place when the river's velocity decreases and it has less energy to transport its load. Many different things can lead to a loss of energy, such as:

- where the river channel bends and water flows more slowly on the inside of the bend
- when a river enters a larger body of water, like a lake or the ocean, and slows down
- when a river floods over large areas of land and temporarily resembles a lake.

A river deposits larger and heavier material first because it requires more energy to carry. A river can usually transport finer and lighter material much further. The finest river sediment is called **alluvium**. It is a very fertile material that is deposited across floodplains following a flood.



Figure 1.3: The river deposits larger pebbles and stones first

River processes working together

Erosion, transportation, and deposition work together to move material along the course of a river. Erosion wears away the landscape and provides material, transportation carries this material downstream, and deposition settles it in new areas. In this way, rivers are continuously redistributing material.

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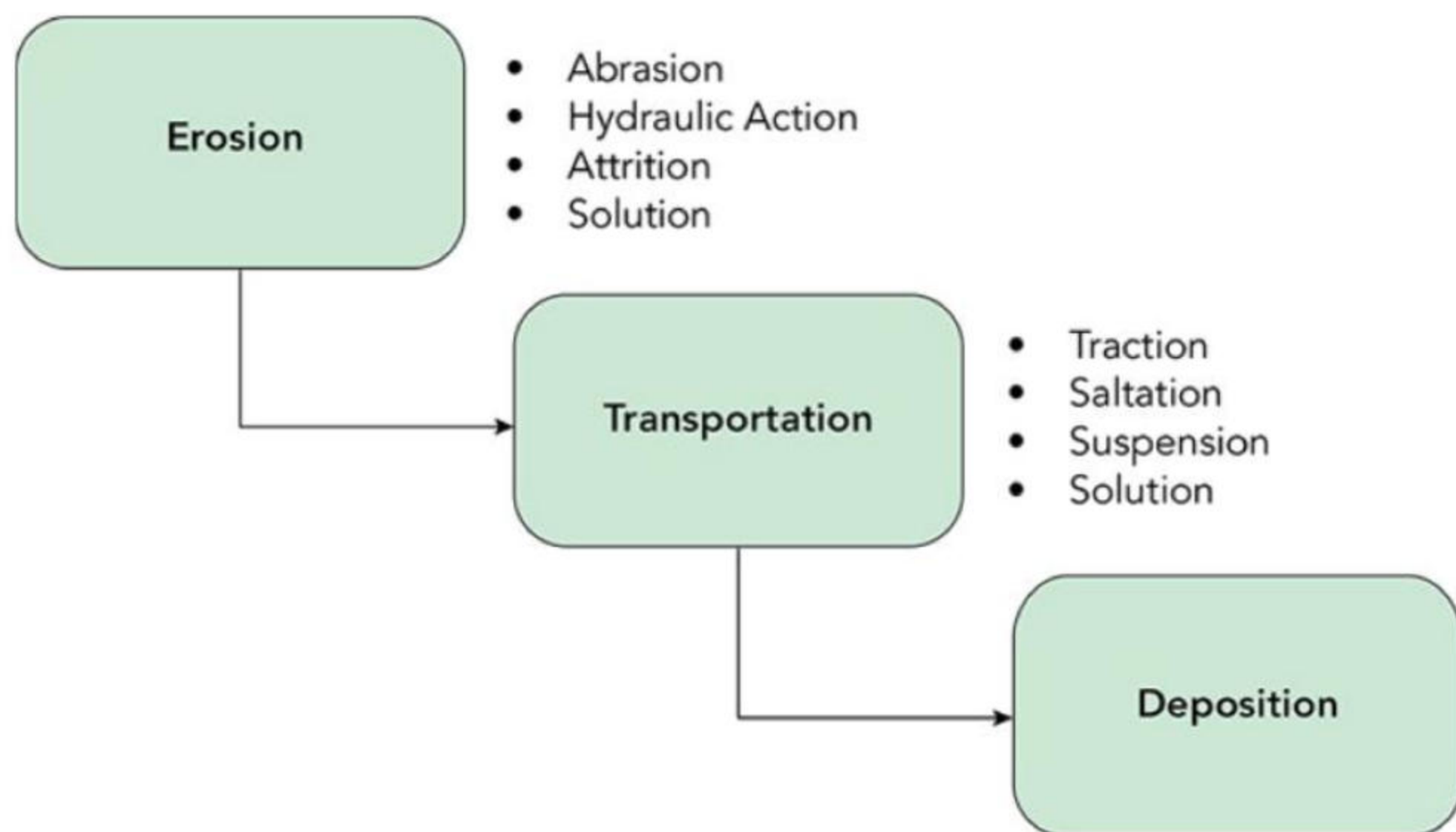


Figure 1.4: River processes work together

TIP

Annotated diagrams can be a useful way of explaining your ideas. Try using diagrams to explain why gravity is an important factor in river processes.

River valleys and the long and cross profiles

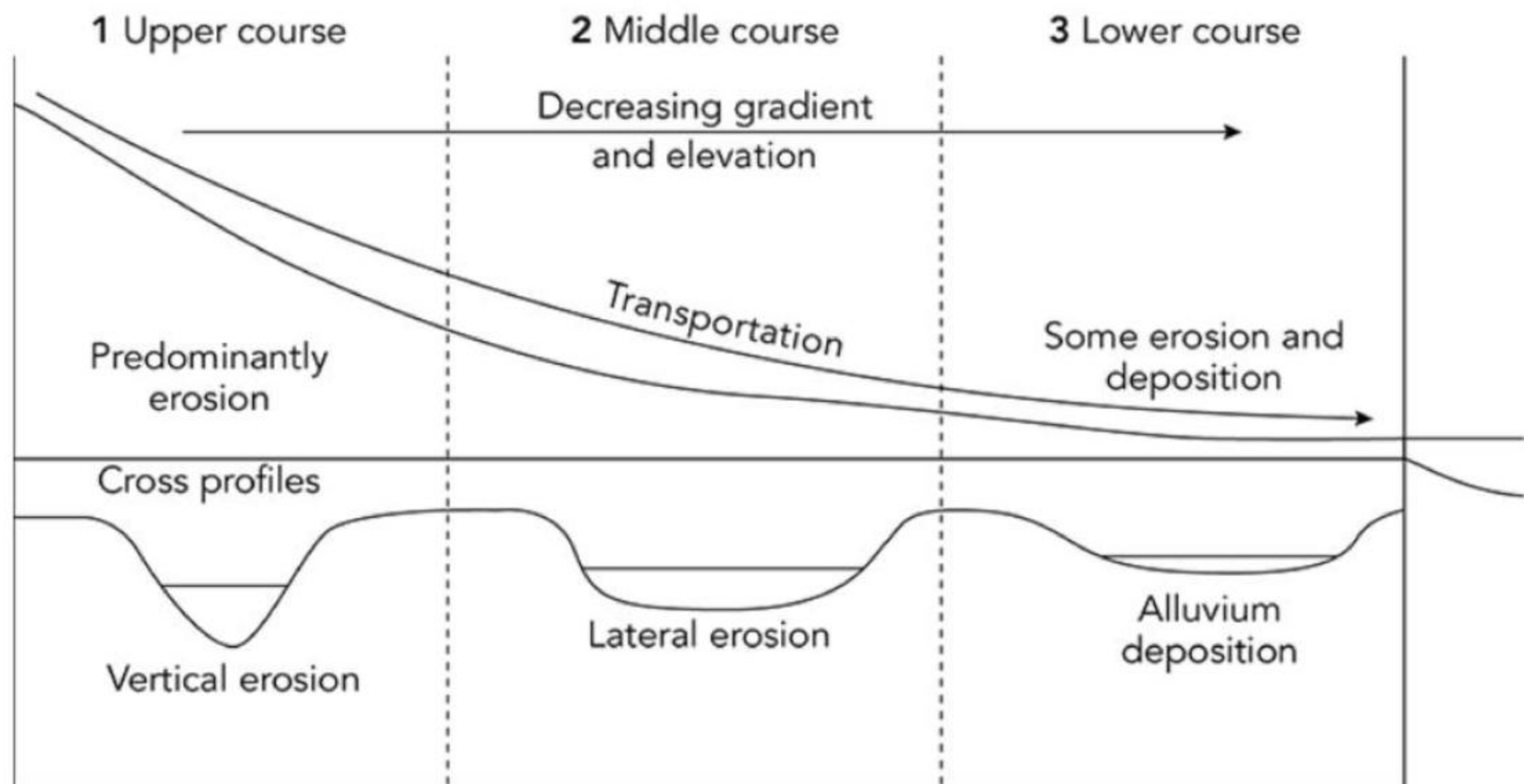


Figure 1.5: River long and cross profiles

River valleys are formed by the continuous processes of flowing water over time. The long profile of a river shows how the river's gradient and **elevation** change along its course. It typically consists of the upper, middle, and lower courses (Figure 1.5). The shape and characteristics of a river valley vary along the long profile:

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- In the upper course, the high elevation leads to vertical, or downwards, erosion. This takes place when the force of gravity, combined with the kinetic energy (energy that is created by movement) of flowing water, enables the river to cut vertically into the landscape. Vertical erosion creates steep V-shaped valleys. As the river flows through V-shaped valleys, interlocking spurs form (Figure 1.6). Interlocking spurs are 'fingers' of harder rock which are harder to erode. This land juts out and the river meanders around it.



Figure 1.6: A V-shaped valley with interlocking spurs of land

- As the river progresses to the middle course, the elevation decreases. Lateral, or sideways, erosion becomes more significant. This is because the river is closer to sea level, but it still has a lot of energy.
- In the lower course, the valley is wide and flat. Some lateral erosion continues. Sediments deposit on the valley floor, creating fertile floodplains.

TIP

Long and cross profiles are easily confused. Try putting an 'a' in front of the words to help remember them. Long profiles are 'a-long' the length of the river. Cross profiles are 'a-cross' the river.

The cross profile of a river refers to the cross-sectional view of the valley. It typically consists of the river channel, the riverbanks, and the land either side of the river. The cross profile in the upper course shows a narrow river channel and V-shaped valley sides. By the lower course, the river channel is wider and deeper, and the valley sides are flat and wide (see Figure 1.5).

TASK 5

- a** The table opposite contains data that students collected on a fieldtrip about the depth of a river. Use the data to **draw a cross-section** of the river channel. The x-axis should be distance across the channel in cm and the y-axis should be depth of the river in cm. The y-axis should have 0 at the top and -15cm at the bottom.
- b** Suggest which course of the river this data has been collected from.
- c** Explain your answer to part **b**.

Distance across the channel (cm)	Depth of the river (cm)
0	0
25	-8
50	-11
75	-14
100	-12
125	-7
150	-5
175	0